



Retail Pricing Analysis & Strategies

Data Analytics Report (2017–2025)

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Dataset overview

This report analyzes monthly retail pricing data for **118 482** products across 11 Canadian provinces from **January 2017** through **December 2025**. Each record includes the product category, product name, pre-tax price (VALUE), whether it's taxable, the total tax rate, and whether the item is considered **Essential** or **Non-Essential**. The wide scope enables long-term trend analysis, cross-province comparisons and category-specific insights.

Note: Provinces were labeled numerically from 1 to 11 in the original dataset for anonymization purposes. This labeling was preserved for analysis and visualization consistency. It does not correspond to Canada's official provincial and territorial structure.

Key observations

- **Average pre-tax price:** The mean price climbed from **\$4.73** in 2017 to **\$6.44** in 2025 ($\approx 36\%$ increase), reflecting a clear inflationary trend.
- **Data distribution:** Essential goods account for roughly 83 % of the records and are about **50 %** more expensive than non-essential goods ($\approx \$5.76$ vs. $\$3.88$).
- **Provincial taxes:** Provinces 2–6 levy the highest average tax rates ($\sim 15\%$), while Province 10 applies only $\sim 5\%$ tax.
- **Product categories:** The dataset contains 14 categories ranging from **Meat & Poultry** to **Infant & Baby Products**. Category-level analysis is crucial for targeted pricing decisions.

Inflation trend

The chart below shows the mean pre-tax price by year. Prices remained fairly stable until 2021 and then rose sharply during 2022–2025. This aligns with broader Canadian inflation trends; Statistics Canada reported that the Consumer Price Index increased **1.9 %** year-over-year in **August 2025**^[1].

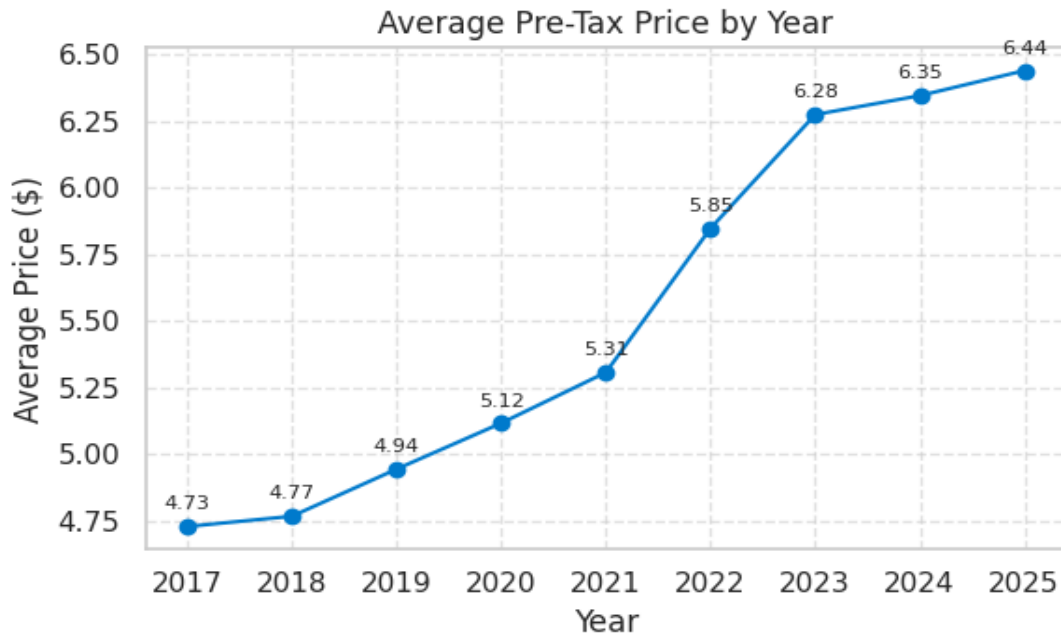


Figure 1: Average pre-tax price trend by year (2017–2025)

What the data reveals

Category profitability & variability

- Most expensive categories:** *Infant & Baby Products* top the list with an average price above **\$16**, followed by *Meat & Poultry* (~\$11) and *Toiletries & Cleaning* (~\$7). The bar-and-error-bar chart below displays each category's average price and variability (standard deviation), with labels positioned immediately after the bar.

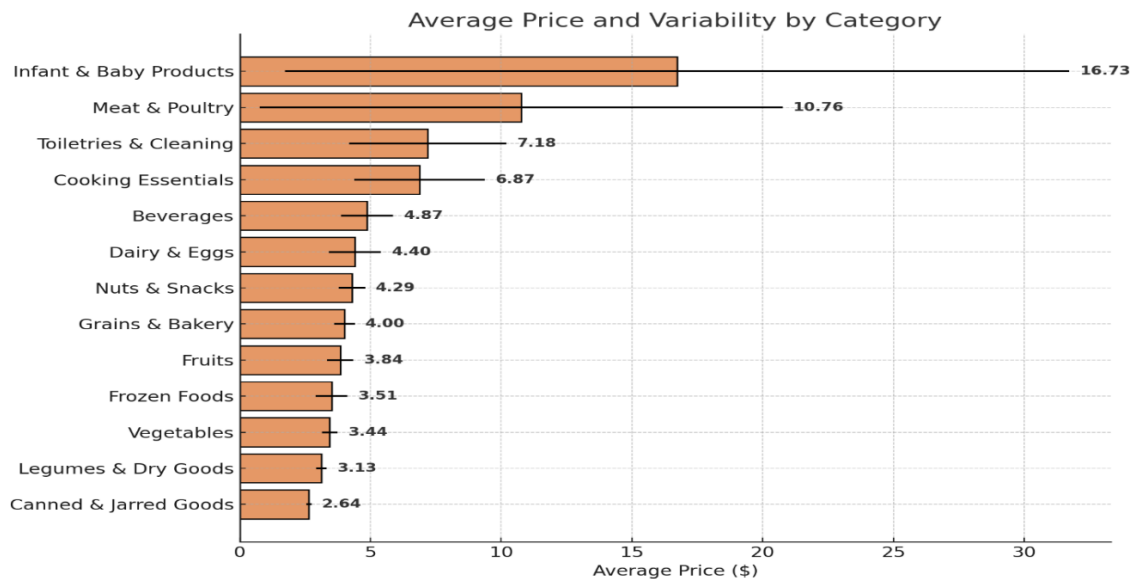


Figure 2: Category-level average price and variability

- **Highest price growth:** *Infant & Baby Products* prices increased by more than **70 %** from 2017 to 2025, while *Cooking Essentials* and *Vegetables* grew by ~50 % and 42 %, respectively.

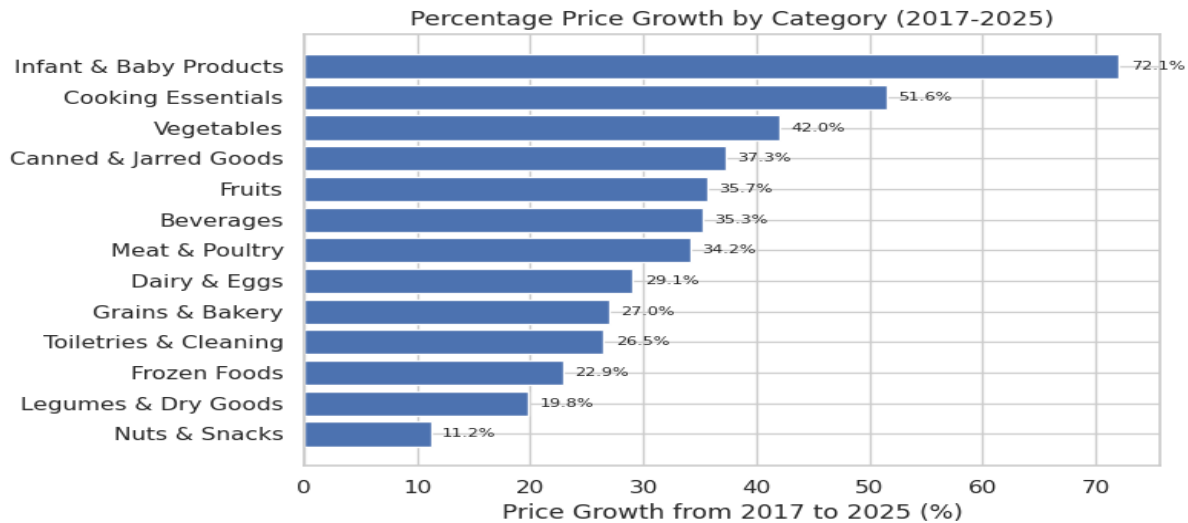


Figure 3: Percentage price growth by category from 2017–2025

- **Variability matters:** *Toiletries & Cleaning* and *Meat & Poultry* show large variability, meaning some items are premium priced while others are low cost. The box plot of the top five categories (below) illustrates this spread—*Infant & Baby Products* have the widest range, reflecting luxury items alongside basic baby supplies.

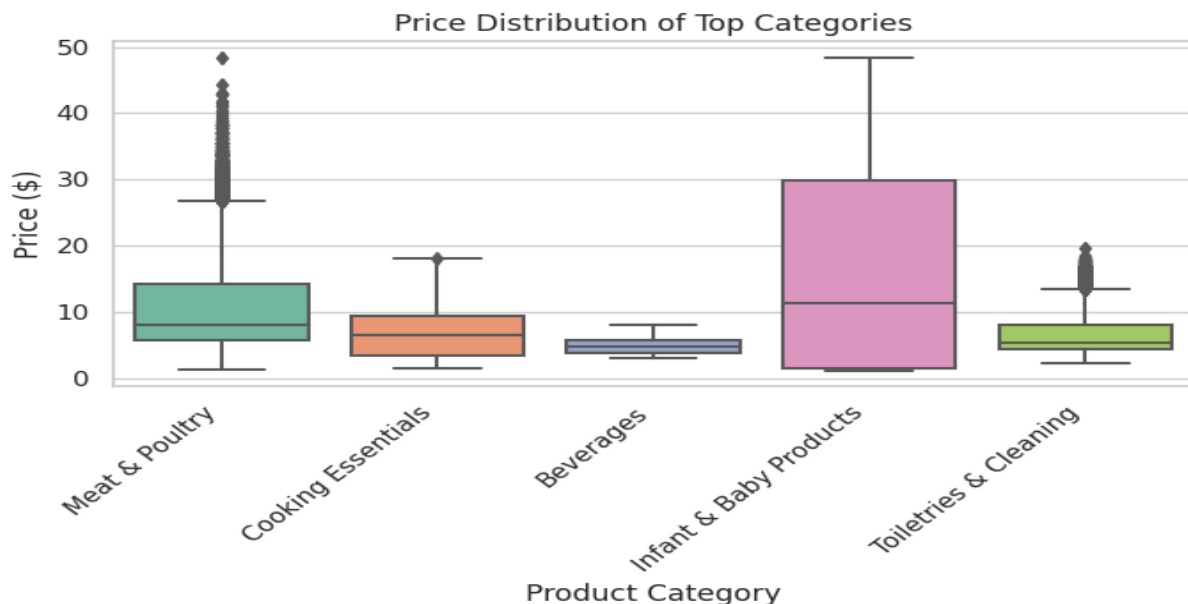


Figure 4: Price distribution for top categories

Market context

- Statistics Canada reports that meat prices rose **7.2 %** year-over-year in August 2025[1], aligning with the notable growth in the *Meat & Poultry* category.
- Overall inflation remains moderate; the CPI increased **1.9 %** year-over-year[1], indicating that some categories are experiencing price increases well above the general inflation rate.
- McKinsey research shows that **even a 1 % improvement in pricing can boost profits by up to 11 %**[2], underscoring the importance of strategic pricing decisions.

Actionable pricing strategies

Drawing from the data analysis and best-practice research, the following strategies are recommended for decision-makers:

1. Implement dynamic pricing on volatile categories

Dynamic pricing adjusts prices in near-real time based on demand, inventory and external factors. An NRS guide notes that this approach can boost profits by **20–25 %**[3]. Focus on categories with high volatility and perishability, such as **Meat & Poultry** and **Infant & Baby Products**. Start by adjusting prices for a few test items at different times of day[3] using POS data to identify peak demand periods[3]. Avoid applying dynamic pricing to staples that consumers expect to remain stable.

2. Adopt value-based pricing for premium categories

Value-based pricing sets prices according to perceived customer value rather than cost. It fosters loyalty and allows premium pricing when the product is differentiated[4]. Apply this strategy to high-quality products such as **premium meat cuts**, **infant formula** and **specialty cooking oils**. Highlight attributes like organic sourcing or brand heritage and use customer surveys to gauge willingness to pay.

3. Utilize bundle pricing to increase basket size

Bundling complementary products into a single package at a better price increases average order value and clears slow-moving inventory[4]. Pair **Cooking Essentials** with **Spices or Beverages** or bundle **Meat & Poultry** items with side dishes. Offer “family dinner kits” or “new baby starter kits” at a slight discount compared with buying items separately, and monitor performance to optimize bundle composition.

4. Create geo-specific and personalized pricing

Adjusting prices based on **location**, **past purchase behavior** or **demographics** helps tailor offers and increase conversion rates[4]. Since provincial tax structures differ, harmonize final prices across regions (by absorbing some tax in high-tax areas) or emphasize low tax in marketing for provinces like **Province 10**. Use

customer-segmentation tools to personalize promotions and provide targeted discounts or loyalty points to offset tax differences.

5. Balance essential vs. non-essential pricing

Essential goods are more price-sensitive and have experienced **37 %** price growth since 2017. Keep these prices stable to maintain customer trust and use them as “loss leaders” to draw traffic. Offset tighter margins with higher pricing on **non-essential** items (luxury snacks, convenience goods) and promote **cross-category bundles** to encourage the purchase of higher-margin items.

6. Leverage predictive analytics

The linear regression model used for this dataset achieved an **R² of 0.94**, indicating that it can reliably predict pre-tax prices. Integrating such predictive models into inventory and pricing tools helps anticipate cost changes and enables data-driven pricing adjustments.

By combining detailed price analytics with evidence-based strategies—dynamic pricing, value-based pricing, bundling, geo-specific pricing and thoughtful segmentation—retailers can respond to inflationary pressures, improve customer satisfaction and maximize profitability.

References

- [1] Statistics Canada. *The Daily — Consumer Price Index, August 2025*.
<https://www150.statcan.gc.ca/n1/daily-quotidien/250916/dq250916a-eng.htm>
- [2] McKinsey & Company. *The power of pricing: How small changes can drive profitability*.
<https://www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/the-power-of-pricing-how-small-changes-can-drive-profitability>
- [3] NRS. *Dynamic Pricing in Retail: Strategies That Work*.
<https://nrsplus.com/blog/dynamic-pricing-retail-strategies/>
- [4] Boostmyshop. *Top 5 Dynamic Pricing Strategies for 2025 to Maximize Profits*.
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